Practical 2: Optics and mineralogy

On successful completion of this practical, you should be able to:

- Describe the relief of a mineral
- Determine whether a mineral is isotropic or anisotropic from its optical properties
- Identify the pleochroic scheme of a mineral
- Identify and describe different types of extinction in anisotropic minerals
- Measure extinction angles
- Estimate birefringence using the Michel-Levy chart
- Predict the optical properties of anisotropic minerals from their lattice parameters and optical symmetry

There are three parts to this practical. Each part should take you about 40 minutes. You can complete them in any order.

How to systematically examine a thin section and make observations.

1. Start by looking at the whole thin section. Start by holding the thin section up to the light to get an overview of the whole section. Make a note of any distinctive features you can see, such as large crystals, distinctive colours, or a fabric.
2. Place the thin section on the microscope stage. Select the lowest available magnification.
3. Look over the whole thin section in PPL. Make a note of any distinctive features you, and any minerals you can identify. Pay particular attention to relief, colour, and cleavage. Be sure to rotate the stage as you proceed, and note any changes in the minerals' appearance.
4. Look over the whole thin section in XPL. Make a note of any distinctive features, and any minerals you can identify. Pay particular attention to interference colours, extinction angles, and twinning.
5. Use the objective lenses to zoom in on any parts of the thin section that are particularly fine-grained, or that you think are particularly interesting.

Part 1 (40 minutes): Thin section 16722, 15, *or* 986

These rocks are **blueschists**. They are metamorphic rocks, formed by high-pressure, low-temperature metamorphism of mafic igneous rocks – but minerals in them are excellent for exploring the optical properties of minerals.

Examine the thin section and find examples of a mineral that is isotropic in cross-polarized light. This is garnet.

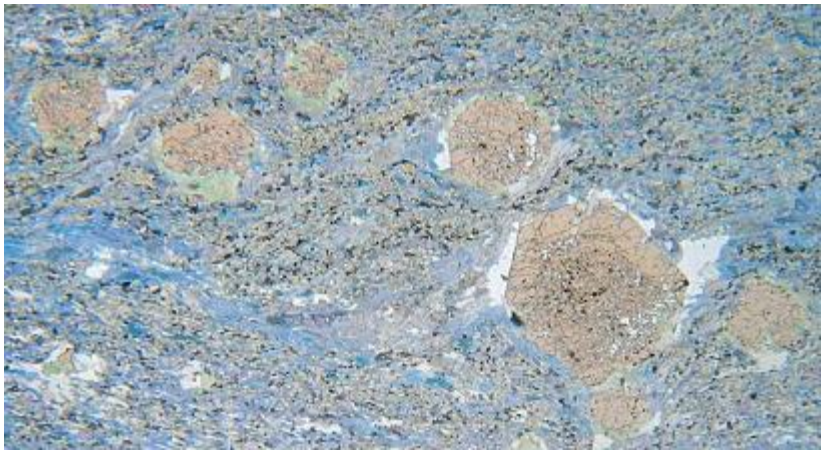
i) What kind of indicatrix does garnet have? Why must this be the case?

It must be an isotropic (spherical) indicatrix. This is the only type of indicatrix with the same refractive index in all directions, and it is the only one that fits with the constraint that the mineral must be in extinction at all times in XPL.

ii) Describe the properties of garnet in plane-polarized light.

The garnets in these thin sections tend to have a pale pink colour, and they are not pleochroic. Garnet has high relief.

iii) Sketch an example of a garnet crystal showing its characteristic crystal habit. This crystal shape and habit is referred to as equant.



This photomicrograph shows the typical equant shape of garnet. Your diagram should capture the rounded to hexagonal shape. Most of the garnets in these thin sections have equant habits, but some are more irregular, or have been sheared during metamorphism.

***The next part of this question is best done with thin sections* 16722 or 15. It will be very difficult using thin section 986 because there is a strong preferred mineral alignment in this section.**

Now find the brightly coloured pleochroic mineral. This is a sodic amphibole called glaucophane. Examine this mineral carefully in PPL and XPL. Make sure you look at different crystals in different parts of the thin section.

iv) What are the three pleochroic colours of glaucophane? From your observations, can you predict the shape of the indicatrix (isotropic, uniaxial, biaxial) in glaucophane?

Blue, lilac, and pale yellow (or colourless). You should be able to find all three colours, although you don't have enough information to work out which is α, which is β and which is γ.

Three pleochroic colours requires three different refractive indices, so glaucophane must have a biaxial indicatrix.

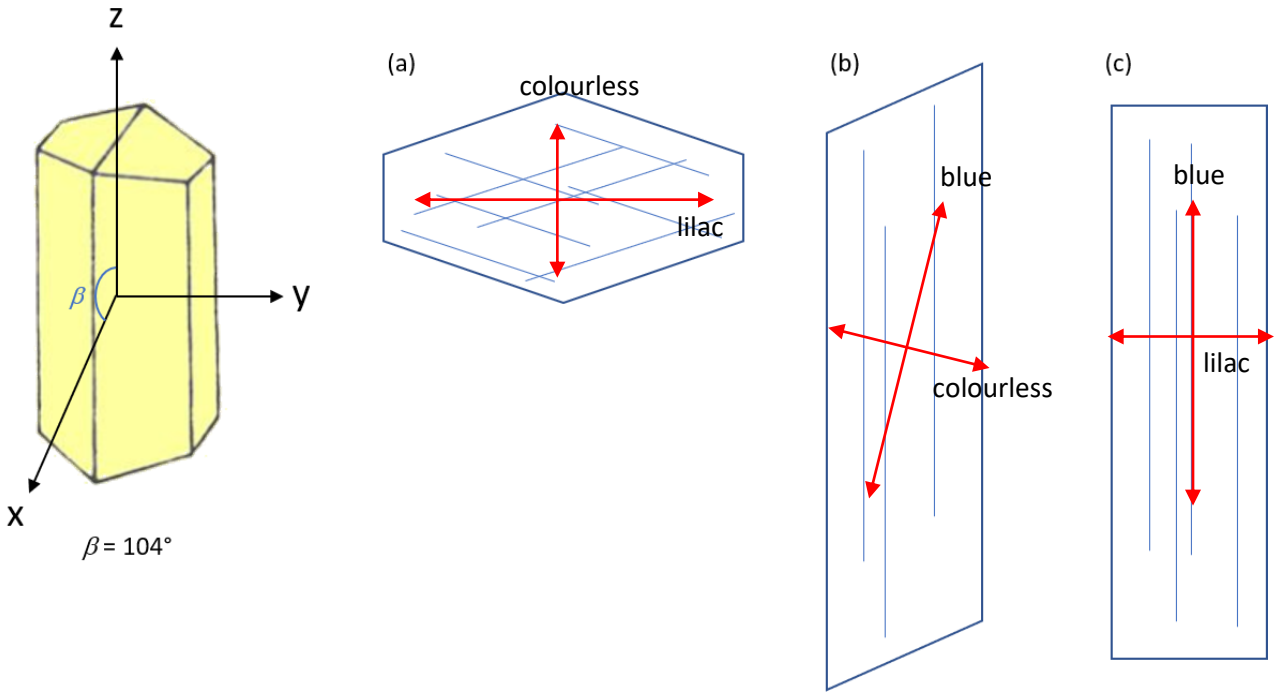
Glaucophane has a prismatic to acicular habit with crystals elongated parallel to the crystallographic z axis. You should be able to find three distinctive sections through the crystal: (a) basal sections with two cleavages (extremely rare in thin section 986); (b) elongate sections with one cleavage and inclined extinction; (c) elongate sections with one cleavage and straight extinction.

v) Add the orientations of the vibration directions in these sections to the diagrams below, and label them with their associated pleochroic colours. Measure the extinction angle for the section with inclined extinction. What kind of extinction do you observe in the basal sections?

The orientations of the vibration directions are marked with red arrows and labelled with their associated colours. The observed colour will depend on which vibration direction is oriented E-W parallel with the polarizer.

The maximum extinction angle in the section with inclined extinction is about 15°. We can do a quick sanity check using the value of lattice angle β: the extinction angle should be equal to $\beta - 90$.

In the basal sections, the vibration directions bisect the cleavage. This is symmetrical extinction.



vi) Which mineral has the refractive index that is most different to the thin section mounting medium: garnet or glaucophane? How do you know?

Garnet, because it has higher relief than glaucophane.

Part 2 (40 minutes): Thin section <u>DS</u> *Dippin Sill, Isle of Arran, Scotland*

This is an **olivine dolerite** (or olivine gabbro). There has been some alteration along the fractures in the olivine in this sample, either to an orange-brown or to a green material. Examine this mineral carefully in PPL and XPL. Make sure that you look at several different crystals in different parts of the thin section.

i) Briefly describe the features of the olivine in PPL. You should include information on the relief, colour, pleochroism (or lack of pleochroism), cleavage, and anything else of interest.

Olivine is colourless with no pleochroism; has quite high relief. Curving fractures are common, and some crystals have faint cleavage traces. Some crystals show evidence of hydrothermal alteration along the fractures and around the crystal edges.

ii) Briefly describe the features of the olivine in XPL. You should include information on the birefringence, type(s) of extinction, and twinning (or lack of twinning).

Olivine has bright second- to third-order interference colours. It does not have twinning. It has straight extinction with respect to the crystal edges, but this can only be seen in crystals with well-developed faces.

iii) What is the highest interference colour that you can find in olivine? What value of birefringence does this indicate? *You will need to use the Michel-Levy chart.*

The highest interference colour should be around third-order blue to green, which corresponds to a birefringence of about 0.038 to 0.040. Your answer will depend on the highest interference colour you identified.

The mineral that is reddish brown in PPL is a pyroxene. Examine this mineral carefully in PPL and XPL. Make sure that you look at several different crystals in different parts of the thin section.

iv) Describe the features of the pyroxene in PPL and XPL.

In PPL, this pyroxene has a reddish brown colour, and there is some faint zoning visible. You may observe some faint pleochroism. It has two cleavages at 90°, although most sections show only one of the cleavage traces. It has moderate to high relief.

In XPL, this pyroxene has first to second-order interference colours – pink-orange and blue-turquoise colours are common. Some of the one-cleavage sections have straight extinction, but others have inclined extinction. Some crystals have twin planes, although this is not a common feature so you may not have seen it.

v) Find a pyroxene crystal with inclined extinction, and measure the extinction angle.

The maximum extinction angle is around 40°.

vi) Is this orthopyroxene (orthorhombic) or clinopyroxene (monoclinic)? Use your observations to support your answer.

It must be clinopyroxene. Orthorhombic minerals can only show straight extinction. Monoclinic crystals can show straight or inclined extinction, depending on whether or not the section contains the y axis.

Notice the mineral that is opaque in PPL. This is an oxide mineral, probably magnetite. Minerals that appear black in PPL are referred to as opaque. Be careful not to confuse opaque minerals with those that only appear black (extinct) in XPL, which are referred to as isotropic.

vii) What is the low-relief mineral? Briefly describe its key features in PPL and XPL.

Plagioclase. In PPL it is colourless and low relief. In XPL it has low first-order interference colours up to white, and multiple lamellar twins that go extinct in different orientations.

viii) If you have time, estimate the modal mineralogy of this rock.

A reasonable estimate would be ~15% olivine, ~35% clinopyroxene, ~40% plagioclase, ~10% opaque oxide. Estimating modal proportions by eye is challenging, so your numbers may be slightly different.

Part 3 (40 minutes): Predicting optical properties using crystallographic information

A hexagonal mineral has a platy habit, forming crystals that are flattened in the direction of the z axis. It has the following mineral parameters:

Unit cell:	a = 9.0 Å	c = 9.2 Å
Refractive indices:	ω = 1.61	ε = 1.56
Pleochroic scheme:	ω = colourless	ε = green
Cleavage:	{001}	

i) What is the shape of the indicatrix?

Uniaxial

ii) Draw sketches of sections cut parallel to (100) and (001) in this mineral (i.e., parallel and perpendicular to the z axis), showing the mineral habit, the cleavage traces, and the orientation of the vibration directions. For each sketch, describe:

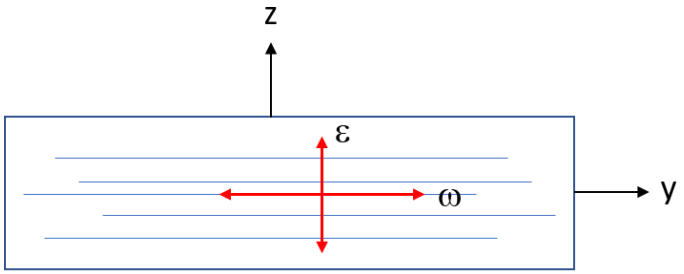
a) The observed pleochroism (if any);

b) The birefringence, and the corresponding interference colour;

c) The type of extinction.

Hint: this question asks for projections showing the crystal habit, which is not necessarily the same as the shape of the unit cell! There is a useful diagram of common crystal habits on p21 of the Practical Reference Handout.

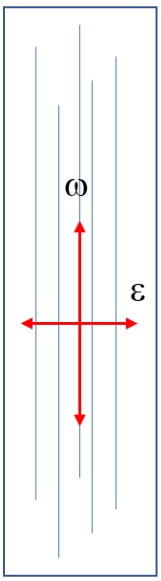
(100)



{001} cleavage traces

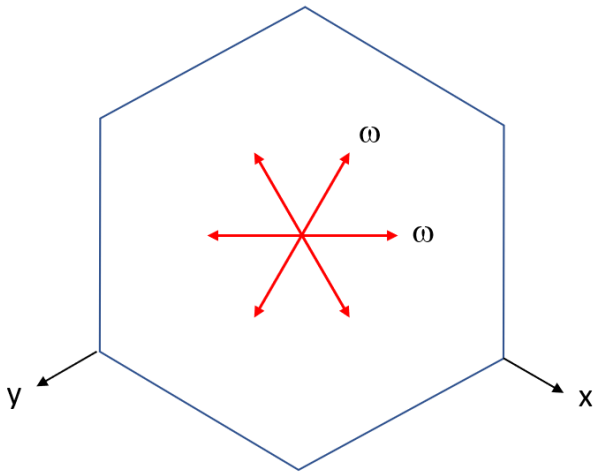
ω is bigger than ε, so represent with a longer arrow.
Colourless in this orientation when ω is aligned E-W

Birefringence = (1.61-1.56) = 0.05
Corresponding interference colour is third-order yellow for a standard 30 µm thin section.



Green in this orientation when ε is aligned E-W

(001)



No visible cleavage traces
In PPL, colourless in all orientations
In XPL, always extinct

Bonus question

If you've finished and would like a challenge, try this question. You can also use it for revision.

Some feldspars have monoclinic symmetry, while others are triclinic.

i) In monoclinic feldspars, one principal axis of the indicatrix must be parallel to the crystallographic *y* axis. Why is this?

The monoclinic feldspar sanidine has the following mineral parameters:

Unit cell:	*a* = 8.6 Å	*b* = 13.0 Å	*c* = 7.2 Å	$\beta = 116°$
Refractive indices:	α = 1.518	β = 1.523	γ = 1.525	
Optical orientation:	β // y	x^$\alpha = 6°$	z^γ = 20°	
Cleavage:	{010} and {001}			
Twinning:	{010}			

ii) Sketch any section through the crystal that contains the y axis, showing the cleavage traces and twin planes, if visible.

iii) Does this section have straight or inclined extinction? *Hint: Extinction should be defined with respect to a crystallographic feature, such as the crystal faces, cleavage, or twin planes.*

iv) If the feldspar was triclinic, would it show straight or inclined extinction in the same section? Why is this?

Solutions available on request! Please email me if you attempted this question and would like feedback on your work.